A Novel Multiplatform Spatiotempoal Data Fusion Approach for Remote Sensing Imagery Based on Parameter Selection

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***Abstract*— Spatiotemporal fusion is an important means to reconstruct the medium spatial resolution remote sensing image series. Presently, many spatiotemporal fusion approaches have been developed and adopted in research on agriculture, ecology, environment, and so on. Although these approaches have achieved remarkable performance in experiments and applications, most of them are designed to fuse all involved bands using the same model with the same parameters, which ignores the band difference. The ignorance may limit the fusion quality for some bands. To address this problem, we propose a novel spatiotemporal data fusion approach based on parameter selection (PSDFA) in this article. The core idea of the newly proposed PSDFA is producing the synthetic image pairs using available data via three means first and then selecting a similar image pair for each band to provide the parameters that are needed for their fusion. The PSDFA can not only be applied in local computers, and its simplified version can also be implemented in Google Earth Engine (GEE), which is a powerful and widely used cloud platform for remote sensing data computing. To test the PSDFA, we conduct two experiments, one in local computers and another in GEE. In local computers, the PSDFA is compared with five state-of-the-art fusion methods on two public Landsat–Moderate Resolution Imaging Spectroradiometer (MODIS) datasets. In GEE, it is used to produce the monthly 30-m image series in two study sites in the USA and compared with another GEE-based fusion approach. The experimental results demonstrate the outstanding performance of the proposed PSDFA in both local computers and GEE.**

***Index Terms*— Band difference, Google Earth Engine (GEE), parameter selection, spatiotemporal fusion.**

I. Introduction

DENSE satellite surface reflectance (SR) image series with medium spatial resolution is usually required by researches on agriculture, ecology, and environment, such as crop yield estimating [1], cropland mapping [2], evapotranspiration estimating [3], surface dynamic mapping [4], soil water content analyzing [5], and suspended particulate matter monitoring [6]. Unfortunately, medium-resolution satellites usually have a long revisit cycle, for example, the revisit cycles of Landsat and SPOT-5 satellites are 16-day and 26-day, respectively, which may be insufficient for the aforementioned requirement. Although some medium spatial resolution satellites with high temporal resolution have been launched, such as the Sentinel-2A and Sentinel-2B, their data are hard to meet that requirement. On the one hand, the cloud coverage seriously reduces the availability of Sentinel-2 images; on the other hand, the Sentinel-2 data coverage is just from 2015 to the present, which can not satisfy the data request for earlier years. To provide dense medium-resolution image series, spatiotemporal data fusion can be adopted, which can be achieved that via fusing the limited medium-resolution images and temporally abundant low spatial resolution images, such as the Moderate Resolution Imaging Spectroradiometer (MODIS) and Sentinel-3 images [7], [8].

In recent years, the spatiotemporal fusion has made great progress. On the one hand, the spatiotemporal fusion has been adopted by various applications to generate the needed data [4], [5], [9], [10]. On the other hand, the development of spatiotemporal fusion approaches is thriving. The core idea of spatiotemporal fusion is combining the spatial details from the images with higher spatial resolution (hereinafter, fine images) and the temporal changes from the low spatial resolution images (hereinafter, coarse images) to predict the missing fine images [11], [12]. Therefore, spatiotemporal fusion approaches usually require at least one fine-coarse image pair from the same date to be the base data, and the coarse image of the predicted date to reconstruct the missing fine image. Most traditional spatiotemporal fusion approaches were designed based on two strategies, which are spatial unmixing (SU) and spatial weighting (SW) [7]. The first SU-based fusion method is called the multisensor multiresolution technique (MMT) [13]. This method assumes all coarse pixels are linearly mixed by fine pixels; thus, it uses unmixing methods to achieve the fusion. First, it classifies the base fine image to acquire endmembers, and then, temporal changes from coarse images are unmixed to reckon the changes of every endmember, which are finally assigned to every base fine pixel to predict the missing fine images. The first SW-based approach is the spatial and temporal adaptive reflectance fusion model (STARFM) [14]. The STARFM assumes all coarse pixels are pure; in this context, the coarse reflectance changes can be added to the base fine image to acquire a transitional prediction. To reduce the error caused by the assumption, the STARFM uses a moving

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window to search the similar pixels for each pixel in the base fine image and establish the weighted relationship between them. Finally, the weighted relationship is utilized to improve the transitional prediction to provide the final prediction. The two approaches can be viewed as prototypes, their SU and SW strategies are inherited by many subsequent spatiotemporal fusion methods. For example, the enhanced spatial and temporal data fusion model [15], the regularized spatial unmixing technique-based spatiotemporal fusion model [16], and the unmixing-based spatiotemporal reflectance fusion model [17] employ the SU strategy. The robust adaptive spatial and temporal fusion model [18], the spatial and temporal nonlocal filter-based fusion model [19], and the enhanced spatial and temporal adaptive reflectance fusion model (ESTARFM) were designed based on the SW strategy. These approaches can be viewed as the enhanced versions of MMT or STARFM. Recently, integrating SW and SU has been a new trend in designing new spatiotemporal fusion approaches. Some hybrid approaches, such as the flexible spatiotemporal data fusion method (FSDAF) [20] and the spatial and temporal reflectance unmixing model [21] have also been developed. Besides these methods, machine learning and deep learning also play roles in the spatiotemporal fusion field, which brought many fusion models [22], [23], such as the nonlocal tensor sparse representation-based semicoupled dictionary learning approach [24], the patch-group compressed sensing-based spatiotemporal fusion model [25] and the pansharpening-based spatiotemporal fusion model [26]. These fusion models can usually achieve remarkable fusion performance.

Though most traditional spatiotemporal fusion approaches have been proven effective in predicting the fine resolution surface changes, including the phenological changes and land cover changes, all of them ignore the band difference. For example, the reliable and adaptive spatiotemporal data fusion method (RASDF) [27] integrates the global SU, local SU and SW to achieve the fusion, which appears outstanding fusion performance in experiments; the virtual image pair-based spatiotemporal fusion approach (VIPSTF) combines multiple available fine-coarse image pairs to provide a virtual image pair, which closer to the missing images of the predicted time. Then, the virtual pair is used as the base data to predict the missing fine images using SU or SW [28], which also exhibits excellent performance. However, the ways they fuse all bands are the same, whose applicability for each band is not discussed. Actually, due to the sensor bias, including different bandwidth coverage and spectral response function, the correlation between different fine and coarse bands are different [29], which may lead to different characteristics in spatiotemporal fusion. Therefore, many fusion approaches usually have advantages in some bands, while in other bands, they are undistinguished. In this context, the spatiotemporal fusion which takes the band difference into account should be discussed. To achieve that, we can gain inspiration from the learning-based fusion methods. Most learning-based fusion models use a uniform model to fuse different bands; however, the parameters of the model for different bands are different due to their total or partial data-driven model training [30], [31], [32], [33]. Therefore, they successfully avoid this problem. It is feasible to design a spatiotemporal fusion approach that can select the best parameters for different bands to enhance their fusion performance.

Since the learning-based fusion models do not suffer from the problem of ignoring band difference, this article focuses on the traditional spatiotemporal fusion approaches. In this article, we develop a new spatiotemporal data fusion approach based on parameter selection (PSDFA). Similar to the learning-based fusion models, the PSDFA fuse different bands using a uniform model; however, whose parameters are different for different bands. Specifically, the PSDFA utilizes three means, which are pixel-level, patch-level, and image-level, respectively, to produce three synthetic fine-coarse image pairs first. Then, each band will select one which is most similar to the predicted data to provide the fusion parameters for it. Finally, all bands will achieve fusion with the best parameters. Plus, the simplified PSDFA can also be implemented in the Google Earth Engine (GEE), which is a widely used cloud platform for remote sensing big data processing [34], [35]. Therefore, the newly proposed is a multiplatform spatiotemporal fusion method.

The main innovative contributions of this article are as follows.

1) The band difference is considered in the spatiotemporal fusion of remote sensing imagery for the first time in this article, which points out a limited vision spot for this field. That may inspire subsequent new methods to pay attention to the performance of different bands.
2) Based on the concern of band difference, a new spatiotemporal fusion approach, i.e., PSDFA, is developed, which is capable of selecting the best fusion parameters for different bands.
3) In addition, the PSDFA is a multiplatform approach, whose simplified version can be applied in GEE. Presently, the spatiotemporal fusion methods in GEE are few, and the PSDFA provides another choice for the researchers in need of data fusion.

The rest of this article is organized as follows. Section II will present the proposed PSDFA in detail. The experimental results will be described in Section III. After that, the analysis and discussion will be presented in Section IV. Finally, the conclusions will be drawn in Section V.

II. Methodology

In this section, we will present the proposed PSDFA. The PSDFA requires two base coarse-fine image pairs, one before and one after the predicted time, to predict the missing fine images. $\mathbf{F}$ and $\mathbf{C}$ will denote the fine and coarse images, respectively, while t_1, t_2, and t_3 refer to their acquisition time. The fine image of time t_2 is the missing one that will be predicted, while the coarse and fine images of time t_1 and t_3 are the base data. Notice that the PSDFA conducts the same operations to each band separately, for which the following description of this method is illustrated based on a single band.

A. Problem Formulation

To achieve the fusion of fine and coarse images, it is necessary to establish a relationship between them. Similar to

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most spatiotemporal fusion approaches, we assume the coarse images are the combination of the upscaled fine images and other sensor bias, which can be represented as

$$\mathbf{C} = G(\mathbf{F}) + \xi \tag{1}$$

where the $G(\cdot)$ is the upscale processing for fine images and ξ is the other sensor bias, such as different bandwidth coverage and spectral response functions. According to this equation, the transformational relationship from coarse images to fine images can not be inferred due to the lack of that from upscaled fine images to fine ones. Thus, we introduce the ratio between fine images and their upscaled ones into this problem, which is

$$\mathbf{R} = \frac{\mathbf{F}}{G(\mathbf{F})}. \tag{2}$$

Notice that the $G(\mathbf{F})$ has been interpolated in this equation, whose size is the same as that of $\mathbf{F}$. Combining (1) and (2), we can acquire the following relationship:

$$\mathbf{F} = (\mathbf{C} - \xi)\mathbf{R}. \tag{3}$$

Similarly, the $\mathbf{C}$ in this equation has been interpolated. Equation (3) provides a way to transfer the coarse images to fine ones; however, there are two parameters that have to be determined first, which are ξ and $\mathbf{R}$. As a matter of fact, for an actual fusion task, the real ξ and $\mathbf{R}$ do not exist; thus, we have to use the ξ and $\mathbf{R}$ from the available coarse–fine image pair as a replacement. It is expected if the image pair is closer to the missing image, the ξ and $\mathbf{R}$ they provided are better for this fusion. The assumptive image pair is called the similar image pair here and denoted as $\mathbf{C}_s$ and $\mathbf{F}_s$, which will provide the parameters

$$\widehat{\xi} = \mathbf{C}_s - G(\mathbf{F}_s) \tag{4}$$

and

$$\widehat{\mathbf{R}} = \frac{\mathbf{F}_s}{G(\mathbf{F}_s)}. \tag{5}$$

Combining (3)–(5), we can acquire the model to predict the $\mathbf{F}_{t_2}$, which is

$$\widehat{\mathbf{F}}_{t_2} = \mathbf{F}_s + \frac{\mathbf{F}_s}{G(\mathbf{F}_s)}\left(\mathbf{C}_{t_2} - \mathbf{C}_s\right). \tag{6}$$

Now, the problems becomes how to acquire the similar image pair and determine the upscale processing, i.e., $G(\cdot)$.

B. Produce and Select the Similar Image Pair

As aforementioned, the higher similarity between the similar pair and the missing image, the higher accuracy of the predicted $\widehat{\mathbf{F}}_{t_2}$ using (6). However, the fine–coarse image pairs of times t_1 and t_3 may not be similar enough. Thus, it is necessary to produce better image pairs. The VIPSTF has proven that the image-level linear regression fitting (LRF) can produce synthetic image pairs which are closer to the missing image. However, the VIPSTF just takes the image-level LRF into account and fuses all bands using the synthetic image pair from that. Considering the band difference, it is necessary to explore more ways for generating synthetic image pairs, and then select the similar pairs for each band. Here, three strategies are adopted to provide synthetic image pairs, which are pixel-level, patch-level, and image-level, respectively.

The pixel-level strategy is the temporally linear interpolation, which can be represented as

$$\mathbf{F}_{s_1} = \mathbf{F}_{t_1} + (t_2 - t_1)\frac{\mathbf{F}_{t_3} - \mathbf{F}_{t_1}}{t_3 - t_1} \tag{7}$$

and

$$\mathbf{C}_{s_1} = \mathbf{C}_{t_1} + (t_2 - t_1)\frac{\mathbf{C}_{t_3} - \mathbf{C}_{t_1}}{t_3 - t_1} \tag{8}$$

where $\mathbf{F}_{s_1}$ and $\mathbf{C}_{s_1}$ are the synthetic fine and coarse images produced by this strategy.

Then, the patch-level strategy is patch-based LRF, which is

$$\mathbf{F}_{s_2}^{p_i} = a_i\mathbf{F}_{t_1}^{p_i} + b_i\mathbf{F}_{t_3}^{p_i} + c_i \tag{9}$$

and

$$\mathbf{C}_{s_2}^{p_i} = a_i\mathbf{C}_{t_1}^{p_i} + b_i\mathbf{C}_{t_3}^{p_i} + c_i \tag{10}$$

where the p_i represents the ith patch of an image, and a_i, b_i, and c_i are the coefficients. The $\mathbf{F}_{s_2}$ and $\mathbf{C}_{s_2}$ are the generated synthetic fine and coarse images using the same coefficients, i.e., a_i, b_i, and c_i, which are determined by LRF on coarse images

$$\left(\widehat{a_i}, \widehat{b_i}, \widehat{c_i}\right) = \underset{(a_i, b_i, c_i)}{\arg\min}\ \Sigma\left\|a_i\mathbf{C}_{t_1}^{p_i} + b_i\mathbf{C}_{t_3}^{p_i} + c_i - \mathbf{C}_{t_2}^{p_i}\right\|^2 \tag{11}$$

where Σ represents the summing of all pixels of a image patch. Notice that there is a problem of determining patch size when using this strategy. The parameters are determined adaptively. First, the patch size is set as the odd multiple of r, i.e., the spatial resolution gap between fine and coarse images, which is then determined by optimizing the following equation:

$$\widehat{\mathrm{ps}} = \underset{\mathrm{ps}}{\arg\min}\ \Sigma_{i=1}^{N}\left\|\mathbf{C}_{s_2}^{p_i} - \mathbf{C}_{t_2}^{p_i}\right\|^2, \quad (\mathrm{ps} = 3r, 5r, \ldots, (2k+1)r \le S) \tag{12}$$

where ps denotes the patch size, k represents the integer, N is the patch number of $\mathbf{C}_{s_2}$ and $\mathbf{C}_{t_2}$, and S is the size of $\mathbf{C}_{s_2}$ and $\mathbf{C}_{t_2}$. Finally, the image-level strategy is the image-level LRF, which is similar to the patch-based LRF but conducted on the whole image. The generated fine and coarse images will be denoted as $\mathbf{F}_{s_3}$ and $\mathbf{C}_{s_3}$.

After the synthetic image pairs are produced, the next step is selecting a similar image pair for each band. Here, we use the pixelwise temporal correlation to determine the similar image pair. Specifically, we first acquire the time series of each coarse pixel, which can be represented

$$T_C(x, y) = \left[\mathbf{C}_{t_1}(x, y), \mathbf{C}_{t_2}(x, y), \mathbf{C}_{t_3}(x, y)\right] \tag{13}$$

where x and y are the coordinates along the horizontal and vertical axes. The T_C is used as the reference to assess the synthetic image pairs. For each synthetic image pair, other two pixelwise time series are prepared, which are

$$T_{s_i}^{C}(x, y) = \left[\mathbf{C}_{t_1}(x, y), \mathbf{C}_{s_i}(x, y), \mathbf{C}_{t_3}(x, y)\right] \tag{14}$$

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and

$$T_{s_i}^F(x, y) = \left[\mathbf{F}_{t_1}^{\text{re}}(x, y), \mathbf{F}_{s_i}^{\text{re}}(x, y), \mathbf{F}_{t_3}^{\text{re}}(x, y)\right] \tag{15}$$

where i is the index of the three synthetic image pairs, and $\mathbf{F}^{\text{re}}$ is the resampled fine images whose size is the same as that of coarse images. The $T_{s_i}^C$ and $T_{s_i}^F$ of every pixel will be used to calculate the correlation coefficients (CCs) with corresponding T_C separately. The two CCs are then added to one, denoted as r_{s_i}. The synthetic image pair, which obtains the highest summed CCs of all pixels, is selected as the similar pair, which is

$$\mathbf{F}_s, \mathbf{C}_s = \underset{(\mathbf{F}_{s_i}, \mathbf{C}_{s_i})}{\arg\max} \ \Sigma_{x=1}^{X} \Sigma_{y=1}^{Y} r_{s_i}(x, y), \quad (i = 1, 2, 3) \tag{16}$$

where X and Y represent the size along the horizontal and vertical axes, respectively.

C. Adaptive Upscaling

After acquiring the similar image pair, the upscale processing, i.e., the $G(\cdot)$, will be determined using it. Most spatiotemporal fusion approaches simplify the upscale processing of fine images as an average aggregation of fine pixels [20]. However, the simplification brings the risk of reducing the accuracy of upscaling. Therefore, we will use an adaptive normalized-flat-Gaussian-kernel to precisely simulate the spatial response between the fine and coarse images, which can be represented as

$$G(\mathbf{F}_s) = \mathbf{F}_s * K_n \tag{17}$$

where the $*$ is the convolutional operation and K_n denotes the normalized-flat-Gaussian-kernel. Due to the point spread effect of satellite sensors, the reflectance of a pixel in remote sensing images will be not only determined by its spatially covered surface, but also influenced by the adjacent surface [36]. In the scenario of upscaling, a coarse pixel is influenced by its spatially covered fine pixels and adjacent ones. Theoretically, the contribution of spatially covered fine pixels is more than the adjacent ones. Thus, the 2-D Gaussian kernel is suitable for simulating this processing. Considering all spatially covered fine pixels by a coarse pixel should has equal contribution, the normal Gaussian kernel is modified to the flat-Gaussian-kernel, supposing its size is $2t + r$, which can be represented as

$$K(x,y) = \begin{cases} \dfrac{1}{r^2}, & |x - x_0| \le \dfrac{r}{2} \ \textbf{and} \ |y - y_0| \le \dfrac{r}{2} \\ \dfrac{1}{r^2}\exp\left(-\dfrac{(x - x_0)^2 + (y - y_0)^2}{2}\right), & \dfrac{r}{2} < |x - x_0| \le t \ \textbf{or} \ \dfrac{r}{2} < |y - y_0| \le t \\ 0, & |x - x_0| > t \ \textbf{or} \ |y - y_0| > t \end{cases} \tag{18}$$

where K is the flat-Gaussian-kernel, and x and y are the coordinates of its horizontal and vertical axes, respectively, while (x_0, y_0) is the coordinates of its center. r refers to the spatial resolution gap between the fine and coarse images. To guarantee the weight sum of the kernel is 1, the K will be normalized before usage, which is the aforementioned K_n. Obviously, there is just a parameter that needs to be determined for the K_n, which is its size, i.e., t. The t is determined using a similar image pair by the following optimization:

$$\widehat{t} = \underset{t}{\arg\min} \ \Sigma \|\mathbf{C}_s - \mathbf{F}_s * K_n\|^2. \tag{19}$$

After that, $G(\cdot)$ is determined.

D. SW Processing

Though we have solved all unknown parameters of (6), an SW processing is necessary. Notice that in (6), the $\mathbf{F}_s$ divide the $G(\mathbf{F}_s)$, which will bring some undesirable problems, such as the aliasing effect or block effect due to their huge spatial resolution gap [14]. The SW has been widely used in spatiotemporal fusion approaches to improve the spatial quality of fused images. Here, we introduce the same SW strategy as that of STARFM (details can be found in [14]) into (6), which will become

$$\widehat{\mathbf{F}_k} = \mathbf{F}_s + \text{SW}\left(\frac{\mathbf{F}_s}{G(\mathbf{F}_s)}(\mathbf{C}_t - \mathbf{C}_s)\right). \tag{20}$$

The $\mathbf{F}_s$ will provide the weighted relationships for the SW processing. The PSDFA is illustrated in Fig. 1.

E. Simplification for Implementation in GEE

Due to the limitations of the GEE platform, the PSDFA can not be applied in GEE directly. Here, we simplify the PSDFA to adapt the GEE in three aspects. First, in GEE, the PSDFA just uses the temporal linear interpolation to produce a similar pair since the GEE is the best at pixelwise processing. Then, the adaptive upscaling is hard to be carried out in GEE; thus, it is replaced with the upscale function provided by the GEE platform. Finally, the SW processing is removed.

III. Experiments

In this section, two experiments, denoted as Experiments I and II, respectively, are conducted to test the proposed PSDFA in different platforms, i.e., local computers and GEE. The experimental information is described as follows in detail.

A. Datasets and Experimental Settings

Experiment I tests the performance of the proposed PSDFA in local computers, in which we compare the proposed PSDFA with five start-of-the-art spatiotemporal fusion approaches on two public datasets. The five approaches are FSDAF, Fit-FC [8], RASDF, ESTARFM, and VIPSTF. Notice that the VIPSTF has two versions, which are VIPSTF-SU and VIPSTF-SW, and the latter has been proven more accurate than the former. Thus, the VIPSTF in this article refers to the VIPSTF-SW. Five widely used indexes are chosen to quantitatively evaluate the fusion results, including the root-mean-square error (RMSE) [37], CC [38], structure similarity (SSIM) [39], erreur relative global adimensionnelle de synthese (ERGAS) [40], and spectral angle mapper (SAM) [41].

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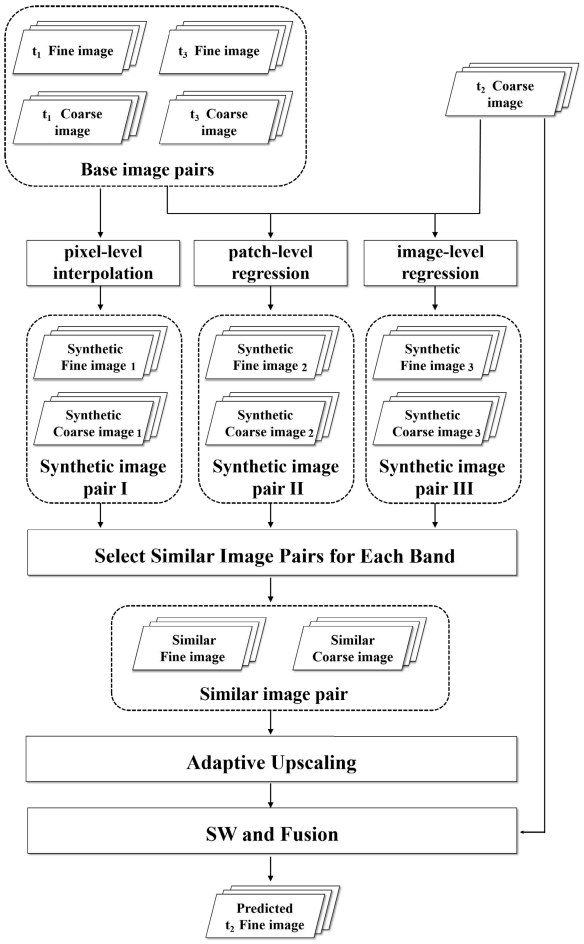


Fig. 1. Flowchart of the proposed PSDFA.

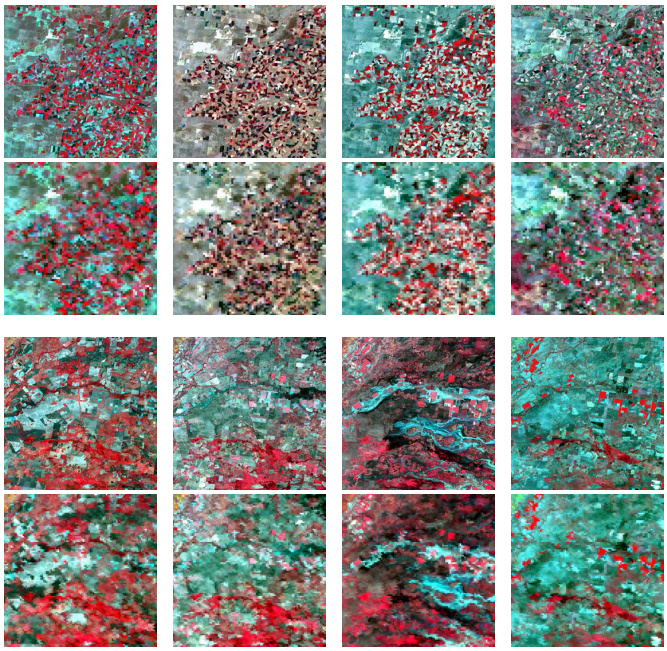


Fig. 2. Exhibition of partial Landsat–MODIS image pairs of CIA and LGC datasets. The first two rows display the Landsat and MODIS images of the CIA dataset, while the bottom two rows show those of the LGC dataset. Three bands (i.e., NIR, red, and green) are, respectively, used as R-G-B bands for demonstration.

Two public Landsat–MODIS SR datasets are selected in the experiments, including the Coleambally Irrigation Area (CIA) dataset and the lower Gwydir catchment (LGC) dataset [42]. The CIA dataset was from the coleambally irrigation area in NSW, Australia. Fifteen Landsat–MODIS image pairs were collected from October 2001 to May 2002. The LGC is located in Northern NSW, Australia [42]. The size of Landsat images in the CIA dataset is 1400 × 1400 pixels while that in the LGC dataset is 2000 × 2000 pixels. All Landsat images have been resampled to 25 m, and all MODIS images are with 500-m spatial resolution. All images contain six bands, including the blue, green, red, near-infrared (NIR), short-wave infrared-1 (SWIR-1), and short-wave infrared-2 (SWIR-2) bands. The two datasets have been widely used for testing the spatiotemporal fusion approaches due to their high quality and challenging fusion scenarios. Specifically, the CIA is a representative agricultural area with high heterogeneity, for which it can test the performance of spatiotemporal fusion approaches in heterogeneous areas. Differently, the LGC suffered from a flood in December 2004. Thus, the LGC dataset is usually used to test the capability of spatiotemporal fusion methods in predicting the landcover changes. Fig. 2 displays partial pairs of CIA and LGC datasets. For the CIA dataset, the second–14th Landsat images will be predicted in the experiment, while their neighbor image pairs will be used as the base data. For example, when we predict the Landsat image of the sixth pair, the fifth and seventh pairs will act as the base data. Similarly, the second–12th Landsat images will be predicted in for the LGC dataset. Notice that the VIPSTF can be implemented when the number of base image pairs is more than two and it is robust to the number of input image pairs [28]. In this experiment, we use all image pairs, except the predicted one, to predict fine images in the CIA and LGC datasets.

In Experiment II, we compare the proposed PSDFA with the highly scalable temporal adaptive reflectance fusion model (HISTARFM) [43], which may be the first spatiotemporal fusion approach in GEE. The HISTARFM uses the Kalman filter and bias-aware Bayesian data assimilation to achieve the fusion of Landsat and MODIS data, which has been proven effective in producing seamless 30-m image series in GEE [43]. Two rectangular areas of the USA are selected in this experiment, which are located in Sacramento, California (SC) and Jackson, Mississippi (JM), respectively. The SC and JM cover 89 × 120 and 94 × 116 km, respectively. The two areas cover many common surface features, including cities, farmlands, mountains, lakes, and reservoirs, for which they are suitable and representative for testing the PSDFA in GEE. In the two areas, the PSDFA and HISTARFM will produce seamless 30-m average SR images of all months in 2020 by fusing Landsat and MODIS data, which then will be compared to the real Landsat images to validate their qualitative and quantitative performance. The used Landsat data include Landsat-7 and Landsat-8 level-2 SR products, while the MODIS data are MOD09A1 and MYD09A1 SR products. Six bands, i.e., blue, green, red, NIR, SWIR-1, and SWIR-2 bands, are involved in this experiment. Similar to the first experiment, we use RMSE, CC, and SSIM to evaluate the fusion results.

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TABLE I
QUANTITATIVE ASSESSMENT OF ALL FUSION RESULTS IN EXPERIMENT I. BOLD INDICATES THE BEST INDICES

	CIA							LGC						
	index	FSDAF	Fit-FC	RASDF	ESTARFM	VIPSTF	PSDFA	index	FSDAF	Fit-FC	RASDF	ESTARFM	VIPSTF	PSDFA
RMSE	2	0.0307	0.0302	0.0315	0.0396	0.0255	**0.0245**	2	0.0208	0.0217	0.0201	0.0253	0.0197	**0.0191**
	3	0.0323	0.0325	0.0331	0.0297	0.0220	**0.0214**	3	0.0321	0.0328	0.0303	0.0207	0.0206	**0.0201**
	4	0.0238	0.0230	0.0236	0.0371	0.0208	**0.0207**	4	0.0247	0.0243	0.0244	0.0354	**0.0220**	0.0233
	5	0.0279	0.0279	0.0280	0.0243	0.0214	**0.0213**	5	0.0246	0.0209	0.0241	0.0177	0.0176	**0.0170**
	6	0.0271	0.0271	0.0264	0.0257	0.0227	**0.0223**	6	0.0246	0.0243	0.0239	0.0267	**0.0238**	**0.0238**
	7	0.0328	0.0320	0.0300	0.0291	0.0275	**0.0272**	7	0.0371	0.0349	0.0358	0.0499	0.0354	**0.0318**
	8	0.0290	0.0283	0.0285	0.0266	**0.0209**	0.0214	8	0.0331	0.0306	0.0336	0.0647	0.0243	**0.0233**
	9	0.0244	0.0243	0.0238	0.0284	0.0233	**0.0232**	9	0.0228	0.0218	0.0221	0.0236	**0.0186**	0.0189
	10	0.0327	0.0318	0.0326	0.0240	0.0189	**0.0169**	10	0.0278	0.0277	0.0271	0.0361	0.0241	**0.0238**
	11	0.0152	0.0209	0.0131	0.0215	0.0153	**0.0127**	11	0.0239	0.0211	0.0231	0.0159	**0.0144**	0.0145
	12	0.0277	0.0383	0.0271	0.0239	**0.0194**	0.0201	12	0.0173	0.0178	0.0167	0.0163	0.0145	**0.0140**
	13	0.0230	0.0227	0.0231	0.0266	0.0199	**0.0198**	-	-	-	-	-	-	-
	14	0.0250	0.0245	0.0254	0.0219	**0.0192**	**0.0192**	-	-	-	-	-	-	-
	average	0.0271	0.0280	0.0266	0.0276	0.0213	**0.0209**	average	0.0262	0.0253	0.0256	0.0302	0.0214	**0.0209**
CC	2	0.8526	0.8510	0.8496	0.8049	0.8942	**0.8976**	2	0.8970	0.8977	0.9049	0.8567	0.9128	**0.9171**
	3	0.8368	0.8345	0.8338	0.8785	0.9236	**0.9266**	3	0.6945	0.7551	0.7265	0.8872	0.9017	**0.9076**
	4	0.9015	0.9043	0.9047	0.8476	0.9224	**0.9238**	4	0.9009	0.9023	0.9048	0.8138	**0.9112**	0.9090
	5	0.8512	0.8457	0.8498	0.9020	0.9159	**0.9182**	5	0.7822	0.8136	0.8001	0.8383	0.8465	**0.8489**
	6	0.8967	0.8939	0.9018	0.9063	0.9273	**0.9293**	6	0.7666	0.7705	0.7871	0.7871	0.7696	**0.7827**
	7	0.8498	0.8551	0.8738	0.8869	0.8951	**0.8980**	7	0.7208	0.7278	0.7289	0.6061	0.7343	**0.7699**
	8	0.8741	0.8717	0.8806	0.9059	**0.9376**	0.9372	8	0.7724	0.8069	0.7588	0.7588	**0.8856**	0.8759
	9	0.9189	0.9206	0.9231	0.8919	0.9306	**0.9318**	9	0.8794	0.8852	0.8886	0.8931	**0.9221**	0.9180
	10	0.8555	0.8613	0.8592	0.9300	0.9589	**0.9725**	10	0.8814	0.8753	0.8877	0.8234	**0.9156**	0.9147
	11	0.9782	0.9332	**0.9867**	0.9355	0.9697	0.9863	11	0.8901	0.8959	0.9010	0.9010	0.9498	**0.9500**
	12	0.7883	0.7197	0.7970	0.8529	**0.8944**	0.8871	12	0.9206	0.9182	0.9273	0.9314	0.9390	**0.9434**
	13	0.8552	0.8563	0.8542	0.8244	0.8832	**0.8840**	-	-	-	-	-	-	-
	14	0.8234	0.8224	0.8205	0.8477	**0.8980**	0.8961	-	-	-	-	-	-	-
	average	0.8679	0.8592	0.8719	0.8780	0.9193	**0.9221**	average	0.8278	0.8408	0.8378	0.8188	0.8807	**0.8852**
SSIM	2	0.8897	0.8883	0.8865	0.8429	0.9221	**0.9260**	2	0.9215	0.9145	0.9275	0.9275	0.9291	**0.9311**
	3	0.8842	0.8753	0.8814	0.9093	0.9454	**0.9478**	3	0.7958	0.8069	0.8178	0.9219	0.9264	**0.9289**
	4	0.9331	0.9345	0.9348	0.8740	0.9482	**0.9486**	4	0.9262	0.9271	0.9297	0.8441	**0.9316**	0.9283
	5	0.8971	0.8914	0.8974	0.9301	0.9410	**0.9423**	5	0.8561	0.8956	0.8646	0.9051	0.9126	**0.9150**
	6	0.9167	0.9136	0.9218	0.9245	0.9414	**0.9418**	6	0.8480	0.8490	0.8614	0.8278	0.8343	**0.8509**
	7	0.8681	0.8660	0.8904	0.9008	0.8990	**0.8997**	7	0.7651	0.7785	0.7772	0.6737	0.7822	**0.8134**
	8	0.8913	0.8809	0.8970	0.9224	**0.9494**	0.9481	8	0.8388	0.8581	0.8287	0.6926	**0.9136**	0.9082
	9	0.9340	0.9310	0.9379	0.9144	**0.9422**	0.9413	9	0.9140	0.9201	0.9203	0.9170	**0.9459**	0.9428
	10	0.8880	0.8920	0.8905	0.9439	0.9665	**0.9770**	10	0.9002	0.8966	0.9073	0.8413	**0.9303**	0.9256
	11	0.9814	0.9475	**0.9880**	0.9491	0.9752	0.9870	11	0.9187	0.9266	0.9255	0.9613	0.9635	**0.9644**
	12	0.8661	0.7912	0.8715	0.9054	**0.9349**	0.9304	12	0.9458	0.9430	0.9504	0.9507	0.9590	**0.9615**
	13	0.9056	0.9004	0.9057	0.8809	**0.9247**	**0.9247**	-	-	-	-	-	-	-
	14	0.8847	0.8818	0.8832	0.9063	**0.9328**	0.9318	-	-	-	-	-	-	-
	average	0.9031	0.8918	0.9066	0.9080	0.9402	**0.9419**	average	0.8755	0.8833	0.8828	0.8562	0.9117	**0.9149**
ERGAS	2	1.1823	1.1507	1.2194	1.5272	0.9845	**0.9402**	2	0.7553	0.7757	0.7348	0.8523	0.6993	**0.6755**
	3	1.1614	1.1593	1.1931	1.0769	0.8089	**0.7892**	3	1.2272	1.3448	1.1698	0.7778	0.7727	**0.7507**
	4	0.9377	0.9017	0.9381	0.9381	**0.7889**	0.7959	4	0.9475	0.9336	0.9388	1.3557	**0.8741**	0.9542
	5	0.9463	0.9574	0.9574	0.8233	**0.7369**	0.7420	5	0.8220	0.7561	0.8090	0.6267	**0.6119**	0.6141
	6	0.8029	0.8028	0.7838	0.7499	0.6569	**0.6523**	6	0.7883	0.7589	0.7663	0.7663	0.7451	**0.7449**
	7	0.9349	0.9349	0.8611	0.8611	0.7914	**0.7816**	7	2.1561	1.9873	2.0639	3.4975	2.0492	**1.8574**
	8	0.9669	0.9134	0.9506	0.9040	**0.7115**	0.7120	8	1.4666	1.3431	1.4941	3.0974	1.0346	**1.0151**
	9	0.8111	0.7963	0.7963	0.9843	**0.7751**	0.7819	9	0.8026	0.7903	0.7778	0.8232	**0.6454**	0.6669
	10	1.1358	1.0926	1.1335	0.7586	0.5972	**0.5451**	10	0.9575	0.9693	0.9343	1.3751	**0.8202**	0.8213
	11	0.4615	0.6844	0.3927	0.6897	0.4883	**0.3862**	11	0.7854	0.7101	0.7591	0.5316	**0.4844**	0.4883
	12	0.9862	1.3908	0.9585	0.8689	**0.6989**	0.7214	12	0.5461	0.5866	0.5260	0.5071	0.4630	**0.4494**
	13	0.9240	0.9304	0.9331	1.0842	**0.8414**	0.8564	-	-	-	-	-	-	-
	14	0.9363	0.8922	0.9481	0.9729	**0.6992**	0.7070	-	-	-	-	-	-	-
	average	0.9375	0.9688	0.9278	0.9769	0.7368	**0.7239**	average	1.0232	0.9960	0.9976	1.2984	0.8364	**0.8216**
SAM	2	0.1200	0.1183	0.1230	0.1044	0.1010	**0.0952**	2	0.0736	0.0707	0.0713	0.0809	0.0669	**0.0651**
	3	0.1237	0.1280	0.1231	0.1038	0.0962	**0.0913**	3	0.1414	0.1310	0.1343	**0.0822**	0.0898	0.0858
	4	0.1039	0.0956	0.0992	0.0928	**0.0874**	0.0886	4	0.1099	0.1030	0.1086	0.1003	**0.0990**	0.1068
	5	0.1005	0.0985	0.0967	0.0816	0.0737	**0.0720**	5	0.0996	0.0879	0.0957	**0.0659**	0.0711	0.0672
	6	0.0772	0.0808	0.0764	0.0779	0.0687	**0.0636**	6	0.0779	0.0751	0.0751	0.0918	**0.0624**	0.0768
	7	0.0885	0.0906	0.0792	0.0760	0.0709	**0.0698**	7	0.2339	0.2138	0.2243	0.2801	0.2211	**0.1991**
	8	0.0914	0.0966	0.0882	0.0733	0.0633	**0.0627**	8	0.1528	0.1446	0.1565	0.1187	0.1243	**0.1103**
	9	0.0754	0.0786	0.0705	0.0828	0.0735	**0.0688**	9	0.0856	0.0893	0.0835	0.0709	**0.0638**	0.0703
	10	0.0956	0.0930	0.0924	0.0707	0.0567	**0.0467**	10	0.0814	0.0809	0.0795	0.0708	0.0756	**0.0671**
	11	0.0496	0.0767	0.0413	0.0645	0.0497	**0.0369**	11	0.0651	0.0682	0.0610	0.0475	0.0459	**0.0452**
	12	0.0930	0.1253	0.0884	0.0780	0.0620	**0.0614**	12	0.0506	0.0537	0.0489	0.0457	0.0429	**0.0406**
	13	0.0794	0.0780	0.0776	0.0737	0.0646	**0.0630**	-	-	-	-	-	-	-
	14	0.0715	0.0717	0.0698	0.0635	0.0542	**0.0537**	-	-	-	-	-	-	-
	average	0.0900	0.0948	0.0866	0.0802	0.0709	**0.0672**	average	0.1065	0.1017	0.1035	0.0959	0.0876	**0.0849**

B. Results of Experiment I

Table I shows the quantitative assessment of six considered fusion methods in all fusion results, in which the RMSE, CC, and SSIM values are averages of all six involved bands. First, it can be seen the newly proposed PSDFA achieves the best in most metrics. Then, it can be found that the VIPSTF is competitive in the fusion results of both CIA and LGC datasets, which outperforms the FSDAF, Fit-FC, RASDF, and ESTARFM. The VIPSTF, which can utilize multiple image pairs to predict the missing fine images, almost achieves the

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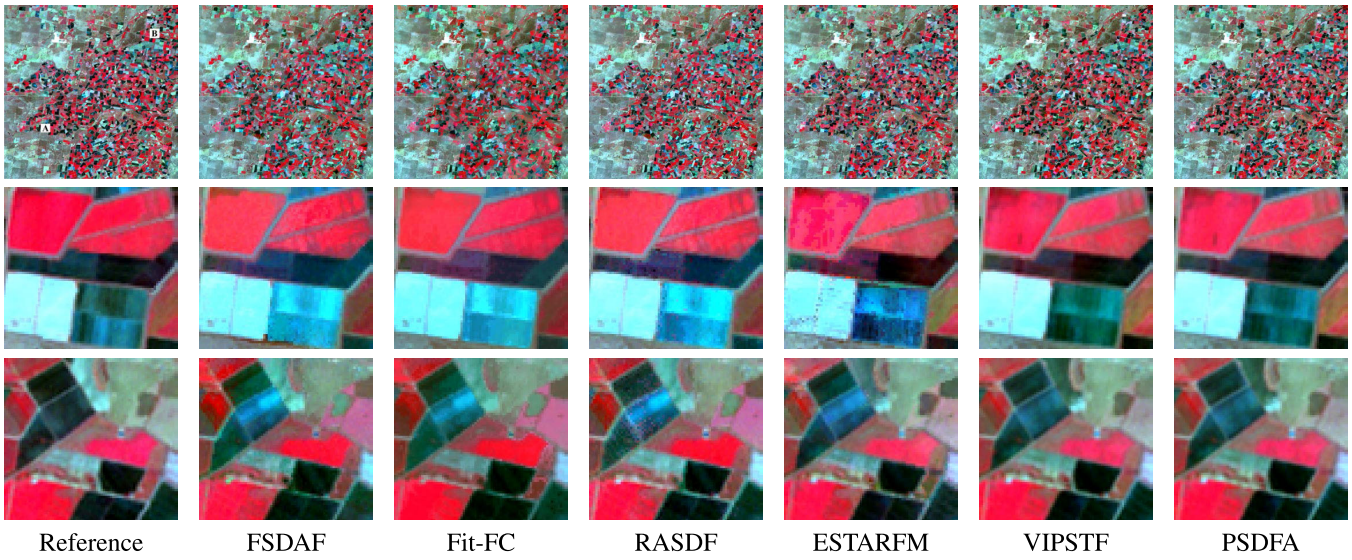


Fig. 3. Fusion results of the third pair of the CIA dataset. The first row shows the fused images, while the last two rows display the images of subarea A and subarea B, respectively.

TABLE II

QUANTITATIVE ASSESSMENT OF FUSION RESULTS OF THE THIRD PAIR OF THE CIA DATASET. BOLD INDICATES THE BEST INDICES

	FSDAF			Fit-FC			RASDF			ESTARFM			VIPSTF			PSDFA		
	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
Blue	0.0108	0.8019	0.9272	0.0105	0.8010	0.9264	0.0110	0.8006	0.9254	0.0093	0.8681	0.9495	0.0089	0.8955	0.9578	**0.0084**	**0.8991**	**0.9604**
Green	0.0138	0.8031	0.9022	0.0140	0.7976	0.8956	0.0143	0.7996	0.8992	0.0109	0.8795	0.9392	0.0100	0.9016	0.9498	**0.0097**	**0.9058**	**0.9518**
Red	0.0209	0.8370	0.8802	0.0204	0.8409	0.8776	0.0216	0.8355	0.8784	0.0162	0.9023	0.9272	0.0140	0.9277	0.9459	**0.0139**	**0.9290**	**0.9465**
NIR	0.0400	0.8946	0.9002	0.0406	0.8925	0.8781	0.0392	0.8970	0.9003	0.0343	0.9213	0.9257	0.0279	0.9492	0.9488	**0.0264**	**0.9543**	**0.9538**
SWIR-1	0.0587	0.8482	0.8510	0.0600	0.8401	0.8379	0.0608	0.8403	0.8456	0.0532	0.8799	0.8846	0.0385	0.9363	0.9370	**0.0378**	**0.9386**	**0.9398**
SWIR-2	0.0498	0.8364	0.8441	0.0495	0.8348	0.8362	0.0515	0.8300	0.8395	0.0546	0.8197	0.8297	0.0326	0.9315	0.9334	**0.0323**	**0.9325**	**0.9342**
ERGAS		1.1614			1.1593			1.1931			1.0769			0.8089			**0.7892**	
SAM		0.1237			0.1280			0.1231			0.1038			0.0962			**0.0913**	

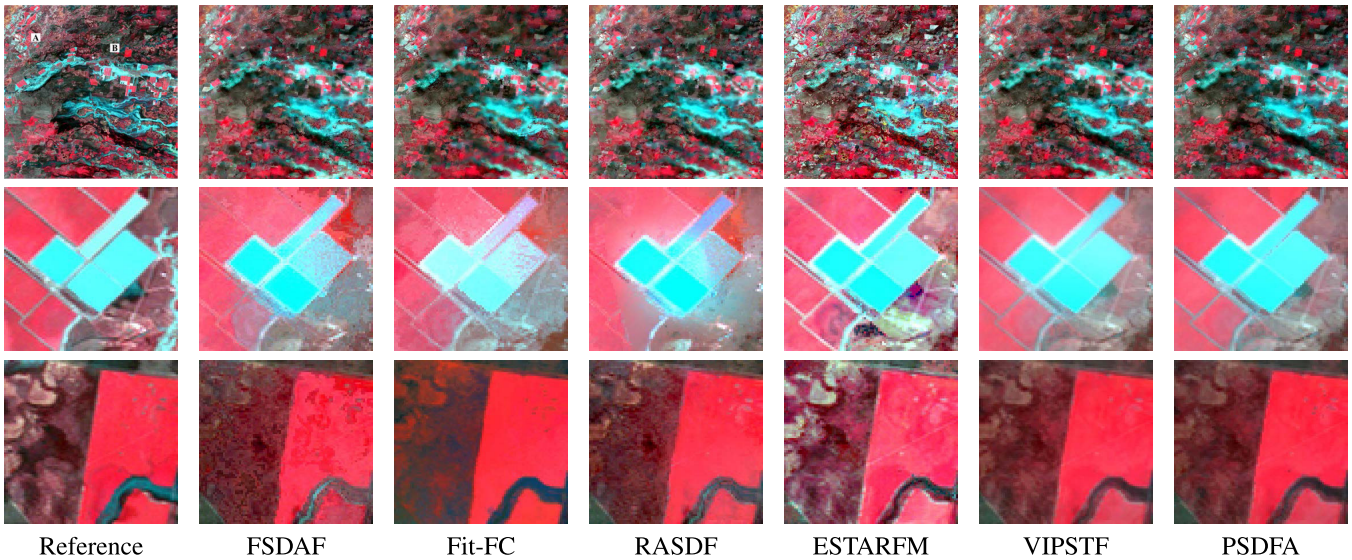


Fig. 4. Fusion results of the seventh pair of the LGC dataset. The first row shows the fused images, while the last two rows display the images of subarea A and subarea B, respectively.

best in half of all fusion results. However, according to the average RMSE, CC, SSIM, ERGAS, and SAM values of all pairs, it can be said that the PSDFA performs better than the VIPSTF, which demonstrates the superior accuracy of the PSDFA. To further contrast the involved methods, we discuss the fusion results of the third pair of the CIA dataset and those of the seventh pair of the LGC dataset in detail. Fig. 3 illustrates the fused images of all six approaches of the third

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TABLE III

QUANTITATIVE ASSESSMENT OF FUSION RESULTS OF THE SEVENTH PAIR OF THE LGC DATASET. BOLD INDICATES THE BEST INDICES

	FSDAF			Fit-FC			RASDF			ESTARFM			VIPSTF			PSDFA		
	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
Blue	0.0161	0.6872	0.8601	0.0155	0.6934	0.8651	0.0155	0.6919	0.8644	0.0159	0.7022	0.8650	0.0160	0.6912	0.8621	**0.0139**	**0.7418**	**0.8861**
Green	0.0221	0.6961	0.8108	0.0216	0.7007	0.8138	0.0217	0.7055	0.8167	0.0224	0.7053	0.8137	0.0219	0.6980	0.8120	**0.0204**	**0.7370**	**0.8334**
Red	0.0267	0.6962	0.7844	0.0261	0.7039	0.7887	0.0264	0.7038	0.7895	0.0275	0.7090	0.7889	0.0260	0.7075	0.7919	**0.0245**	**0.7428**	**0.8125**
NIR	0.0415	0.8001	0.7985	0.0418	0.7931	0.7939	0.0410	0.8022	0.7994	0.0438	0.7564	0.7790	0.0387	0.8323	0.8325	**0.0353**	**0.8589**	**0.8498**
SWIR-1	0.0643	0.7314	0.6911	0.0593	0.7372	0.7063	0.0616	0.7414	0.7098	0.0931	0.4691	0.4876	0.0627	0.7397	0.7020	**0.0568**	**0.7665**	**0.7302**
SWIR-2	0.0517	0.7140	0.6460	0.0450	0.7385	0.7035	0.0485	0.7288	0.6832	0.0970	0.2945	0.3081	0.0472	0.7372	0.6927	**0.0425**	**0.7728**	**0.7318**
ERGAS		2.1561			1.9873			2.0639			3.4975			2.0492			**1.8574**	
SAM		0.2339			0.2138			0.2243			0.2801			0.2211			**0.1991**	

TABLE IV

QUANTITATIVE ASSESSMENT OF ALL INVOLVED FUSION APPROACHES IN DIFFERENT BANDS IN THE CIA DATASET. BOLD INDICATES THE BEST INDICES

	FSDAF			Fit-FC			RASDF			ESTARFM			VIPSTF			PSDFA		
	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
Blue	0.0112	0.8508	0.9368	0.0115	0.8409	0.9310	0.0112	0.8532	0.9376	0.0111	0.8497	0.9351	0.0089	0.9031	0.9592	**0.0088**	**0.9056**	**0.9602**
Green	0.0140	0.8420	0.9152	0.0145	0.8269	0.9040	0.0138	0.8449	0.9172	0.0122	0.8828	0.9371	0.0110	0.9039	0.9474	**0.0109**	**0.9057**	**0.9484**
Red	0.0214	0.8663	0.8961	0.0221	0.8559	0.8839	0.0208	0.8725	0.9021	0.0191	0.8997	0.9221	0.0166	0.9223	0.9385	**0.0163**	**0.9286**	**0.9394**
NIR	0.0375	0.8552	0.8662	0.0379	0.8506	0.8490	0.0364	0.8643	0.8744	0.0337	0.8911	0.8993	0.0286	0.9160	0.9199	**0.0281**	**0.9225**	**0.9224**
SWIR-1	0.0421	0.8939	0.8989	0.0439	0.8865	0.8870	0.0417	0.8953	0.9009	0.0450	0.8854	0.8902	0.0336	0.9336	0.9360	**0.0331**	**0.9390**	**0.9383**
SWIR-2	0.0363	0.8990	0.9052	0.0378	0.8944	0.8962	0.0358	0.9012	0.9077	0.0443	0.8596	0.8642	0.0289	0.9369	0.9403	**0.0276**	**0.9328**	**0.9442**
STDEV	0.0132	0.0235	0.0233	0.0137	0.0263	0.0269	0.0130	0.0226	0.0208	0.0155	0.0193	0.0286	0.0104	0.0144	0.0130	**0.0101**	**0.0140**	**0.0125**

TABLE V

QUANTITATIVE ASSESSMENT OF ALL INVOLVED FUSION APPROACHES IN DIFFERENT BANDS IN THE LGC DATASET. BOLD INDICATES THE BEST INDICES

	FSDAF			Fit-FC			RASDF			ESTARFM			VIPSTF			PSDFA		
	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
Blue	0.0122	0.8003	0.9173	0.0132	0.8094	0.9173	0.0118	0.8122	0.9216	0.0103	0.8567	0.9396	0.0103	0.8652	0.9436	**0.0101**	**0.8728**	**0.9445**
Green	0.0149	0.8116	0.8985	0.0141	0.8316	0.9097	0.0147	0.8186	0.9008	0.0129	0.8565	0.9216	0.0123	0.8654	0.9220	**0.0121**	**0.8711**	**0.9296**
Red	0.0193	0.8234	0.8794	0.0187	0.8413	0.8896	0.0190	0.8319	0.8845	0.0172	0.8650	0.9066	0.0158	0.8779	0.9181	**0.0156**	**0.8829**	**0.9185**
NIR	0.0370	0.8492	0.8622	0.0341	0.8718	0.8829	0.0358	0.8620	0.8736	0.0327	0.8719	0.8879	0.0290	0.9033	0.9109	**0.0285**	**0.9088**	**0.9149**
SWIR-1	0.0381	0.8416	0.8495	0.0369	0.8402	0.8456	0.0373	0.8497	0.8579	0.0516	0.7548	0.7661	0.0319	0.8808	0.8836	**0.0313**	**0.8843**	**0.8842**
SWIR-2	0.0360	0.8406	0.8460	0.0346	0.8503	0.8545	0.0347	0.8523	0.8582	0.0566	0.7078	0.7154	0.0289	0.8915	0.8919	**0.0278**	**0.8912**	**0.8976**
STDEV	0.0121	0.0192	0.0283	0.0111	0.0206	0.0288	0.0116	0.0200	0.0251	0.0201	0.0696	0.0924	0.0098	0.0149	**0.0217**	**0.0093**	**0.0138**	**0.0127**

pair of the CIA dataset, in which the first row shows the fused images while the last two rows display those of two subareas. The CIA is quite heterogeneous, which can be seen from Fig. 3. Generally speaking, it can be seen that all fused images are similar to the reference image from the first row of Fig. 3. However, the two subareas reveal the difference. According to subarea A, it can be said the VIPSTF and the PSDFA achieve the best visual performance, whose results are quite close to the reference image, so do subarea B. In addition, Table II shows the quantitative assessment of the fusion results of the fusion results, from which it can be learned the PSDFA is more accurate than the other five approaches. Similarly, fusion results of the seventh pair of the LGC dataset are exhibited in Fig. 4, which exhibits the serious land cover changes due to the flood. From the first row, it can be seen that all six approaches successfully capture the land cover changes. However, according to the two subareas, the VIPSTF and PSDFA are still the best two approaches to reconstruct the fine image. Then, the quantitative assessment for the fusion results in this pair is shown in Table III, which further demonstrates the best one is the PSDFA.

For the purpose of evaluating the performance of all six involved fusion approaches in different bands, we average their RMSE, CC, and SSIM values of all fused images in the CIA and LGC datasets for each band, and calculate standard deviation (STDEV) of each index, which are shown in Tables IV and V, respectively. According to Tables IV and V, two points should be noticed. One is the proposed PSDFA that is outstanding in all six bands in both CIA and LGC datasets, which demonstrates its high accuracy in all bands. The other is the PSDFA that is the most balanced approach for different bands, which can be found via the smallest STDEV values of the PSDFA. In contrast, the FSDAF, Fit-FC, RASDF, and ESTARFM appear less balanced than the PSDFA. As for the VIPSTF, its balance for different bands is close to the PSDFA; however, it is still outperformed in this aspect. From this experiment, we can learn the PSDFA can not only achieve excellent fusion performance and accuracy, but also balance that for all bands. This is mainly from the parameter selection mechanism, which provides the best parameters for all bands, ensuring the high fusion quality of them. When all bands are well-fused, the balance is maintained.

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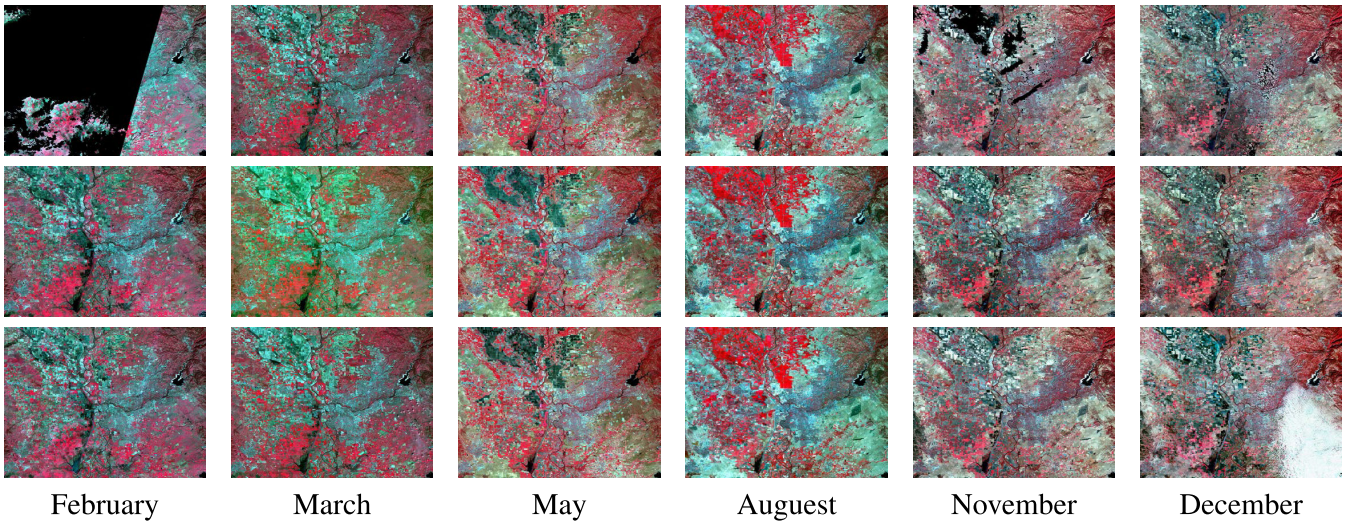


Fig. 5. Partial comparisons among the (first row) real Landsat images, (second row) fused images of HISTARFM, and (third row) PSDFA in the SC.

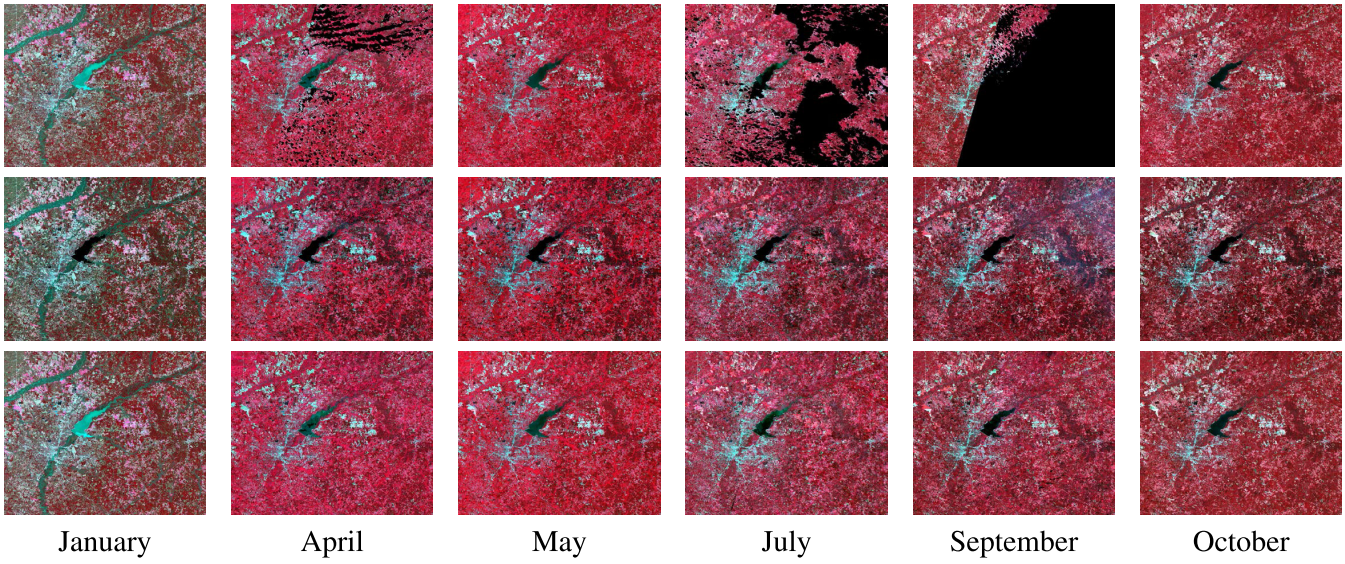


Fig. 6. Partial comparisons among the (first row) real Landsat images, (second row) fused images of HISTARFM, and (third row) PSDFA in the JM.

C. Results of Experiment II

Figs. 5 and 6 exhibit the comparison among the real Landsat images, fused images of HISTARFM and PSDFA of SC and JM, respectively. First, it can be seen that the two approaches are both successful in filling the loss of Landsat images by fusing them with MODIS images, which can be proven by the images of February and November in SC and those of April, July, and September in JM. Compared to the HISTARFM, the proposed PSDFA appears more robust. For example, the fused image of HISTARFM of May in SC and that of September in JM appear visual distortion, which is avoided by the PSDFA. Plus, the fused images of PSDFA are more closer to the real Landsat images, which can be demonstrated by the images of January, May, and October in JM. Figs. 7 and 8 provide subareas comparison, which further demonstrates the robustness of the PSDFA. On the other hand, it is also necessary to discuss the limitations of the proposed PSDFA in GEE. The PSDFA does not perform better than HISTARFM in December in SC since there is the massive cloud in its fused image. The reason why cloud exists in the fused images is the quality assessment (QA) bands of some Landsat images do not flag all cloud successfully. Therefore, the performance of PSDFA is seriously limited by the accuracy of the QA bands of fine images. Differently, the HISTARFM are not affected by that seriously. Thus, when using the PSDFA in GEE, it is recommended to adopt better cloud removal approaches in the preprocessing of fine images. The quantitative assessment of all fused images is illustrated in Fig. 9, from which we can learn that the PSDFA outperforms the HISTARFM in 11 months in SC and all months in JM. Fig. 9 indicates the high accuracy of the proposed PSDFA again.

In summary, compared to FSDAF, Fit-FC, RASDF, ESTARFM, and VIPSTF in local computers, the PSDFA can achieve impressive qualitative and quantitative fusion performance. And the PSDFA is balanced for different bands. In GEE, the PSDFA still has remarkably qualitative and quantitative fusion performance; however, its performance is limited by the accuracy of cloud removal of fine images. In addition, this experiment also points out the importance of the quality of images involved in data fusion for the fusion

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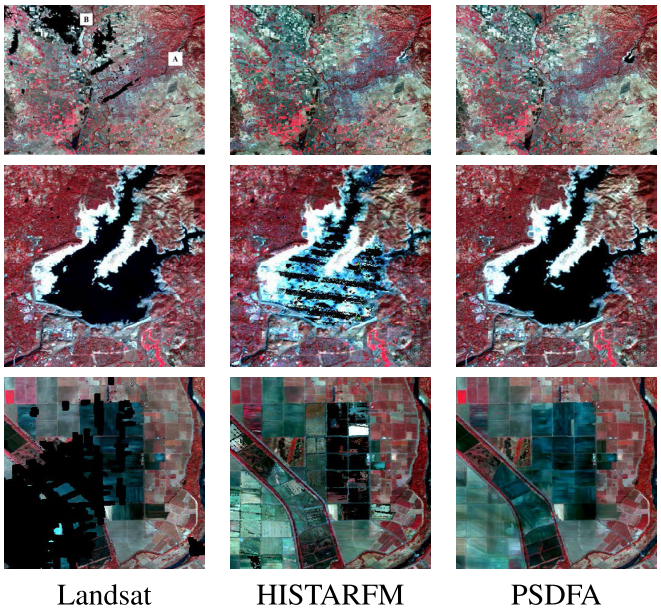


Fig. 7. Fusion results in the SC, November 2020. The first row shows the whole area images, while the last two rows display the images of subarea A and subarea B, respectively.

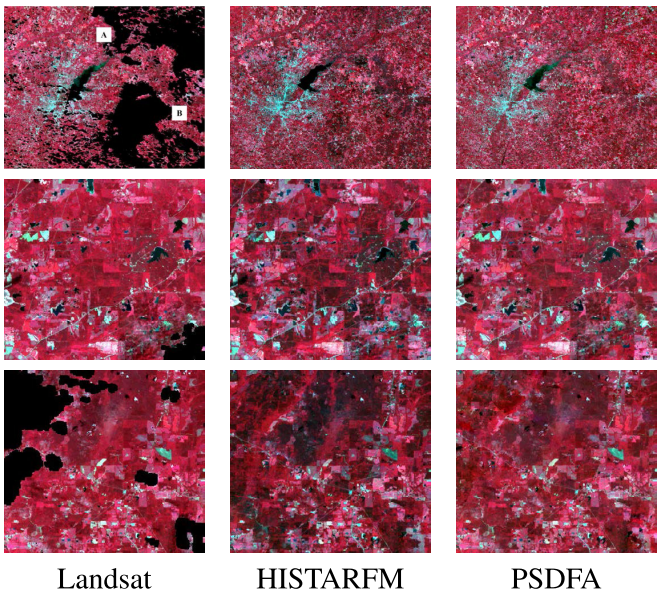


Fig. 8. Fusion results in the JM, July 2020. The first row shows the whole area images, while the last two rows display the images of subarea A and subarea B, respectively.

results. Sometimes, the systematically geometric correction and QA band-based cloud removal may be insufficient. In this case, relative geometric correction and better cloud removal approaches, such as machine learning-based ones, are recommended to be adopted to process the images before data fusion.

IV. Analysis and Discussion

Though two experiments have proven the remarkable fusion performance of the proposed PSDFA, further analysis is necessary. In this section, we will discuss the contributions of the two unique and important components of the PSDFA, which are similar image pair selection (SPS) and adaptive upscaling (AUP). Then, the limitations of this approach will be discussed.

A. Contribution of the SPS

The SPS is an important component of the proposed PSDFA. To illustrate its contribution, we select the sixth and seventh pairs of the CIA and LGC datasets, respectively, to compare the fusion results of PSDFA using different synthetic image pairs. The synthetic image pairs mentioned in Section II will be denoted as $\mathbf{P}_{s_1}$, $\mathbf{P}_{s_2}$, and $\mathbf{P}_{s_3}$ here. Notice the $\mathbf{P}_{s_1}$ means the collection of $\mathbf{F}_{s_1}$ and $\mathbf{C}_{s_1}$, so do $\mathbf{P}_{s_2}$ and $\mathbf{P}_{s_3}$. Four scenarios are taken into account, including just using the $\mathbf{P}_{s_1}$, just using $\mathbf{P}_{s_2}$, just using $\mathbf{P}_{s_3}$, and using the SPS. The fusion results are shown in Table VI, which reveals the effect of SPS. First, it can be seen different synthetic image pairs provide different fusion performances for different bands. In the sixth pair of the CIA dataset, $\mathbf{P}_{s_1}$ and $\mathbf{P}_{s_2}$ are the best for NIR and blue bands, respectively, while $\mathbf{P}_{s_3}$ wins the best in other three bands. Differently, $\mathbf{P}_{s_3}$ performs best in the first three bands in the seventh pair of the CIA dataset, while $\mathbf{P}_{s_1}$ and $\mathbf{P}_{s_2}$ achieve better performance in the last three bands. Therefore, it can be said that there is a large uncertainty in the performance of different synthetic image pairs for different bands in different data. Fortunately, the SPS is capable of determining the best synthetic image pairs for different bands. From Table VI, we can learn the SPS correctly selects the best synthetic image pairs for all bands. On the other hand, it also indicates the necessity of considering the band difference in spatiotemporal fusion. Notice that the fused images using SPS are slightly different from those using a single synthetic image pair, which is caused by the SW procedure. The SW procedure utilizes the class maps from fine images to determine the similar pixels in a moving window, which are different in the cases of using SPS and a single synthetic image pair. In addition, we calculate the mean RMSE, CC, SSIM, ERGAS, and SAM values of the PSDFA using $\mathbf{P}_{s_1}$, $\mathbf{P}_{s_2}$, $\mathbf{P}_{s_3}$, and SPS, which are recorded in Table VII. Table VII quantitatively shows the accuracy improvement brought by the SPS in the CIA and LGC datasets. It further demonstrates the contribution of SPS.

B. Contribution of the AUP

AUP is another unique component of the PSDFA. According to (6), it can be learned that AUP works for providing transferring coefficients between the coarse and fine images. To investigate its contribution, we compare it with the other two strategies. One is replacing the AUP with average pixel aggregation (APA), which is widely used for upscaling in existing spatiotemporal fusion approaches, and the other is removing the upscaling procedure, which can be represented as

$$\widehat{\mathbf{F}_k} = \mathbf{F}_s + \mathrm{SW}(\mathbf{C}_t - \mathbf{C}_s). \tag{21}$$

Tables VIII and IX show the quantitative performance of the fusion results of a different PSDFA, in which the PSDFA removing upscaling procedure can be viewed as the reference to judge the AUP and APA bring improvement or impairment for the fusion. First, the AUP successfully improves the fused

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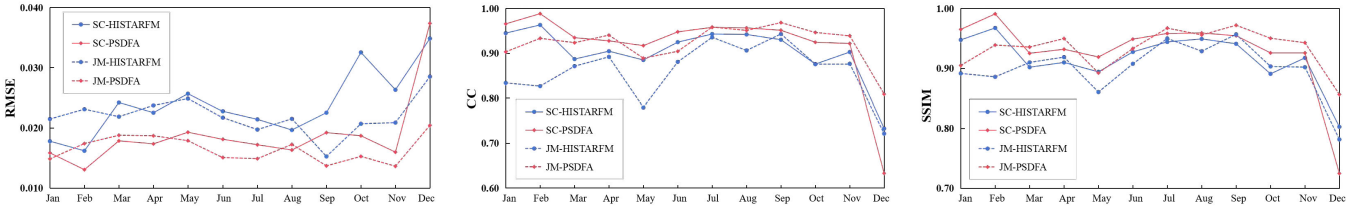


Fig. 9. Quantitative assessment of Experiment II.

TABLE VI

QUANTITATIVE ASSESSMENT OF DIFFERENT FUSED IMAGES OF THE PSDFA IN DIFFERENT FUSION SCENARIOS OF THE SIXTH AND SEVENTH PAIRS OF THE CIA AND LGC DATASETS, RESPECTIVELY

		CIA			LGC		
		RMSE	CC	SSIM	RMSE	CC	SSIM
Blue	$\mathbf{P}_{s_1}$	0.0123	0.9094	0.9499	0.0154	0.7237	0.8753
	$\mathbf{P}_{s_2}$	0.0114	0.9163	0.9539	0.0151	0.7054	0.8702
	$\mathbf{P}_{s_3}$	0.0116	0.9204	0.9562	0.0139	0.7411	0.8858
	SPS	0.0112	0.9204	0.9564	0.0139	0.7418	0.8861
Green	$\mathbf{P}_{s_1}$	0.0138	0.9109	0.9380	0.0213	0.7262	0.8283
	$\mathbf{P}_{s_2}$	0.0133	0.9135	0.9396	0.0214	0.7111	0.8196
	$\mathbf{P}_{s_3}$	0.0132	0.9177	0.9443	0.0204	0.7368	0.8333
	SPS	0.0128	0.9183	0.9471	0.0204	0.7370	0.8334
Red	$\mathbf{P}_{s_1}$	0.0197	0.9357	0.9384	0.0258	0.7322	0.8082
	$\mathbf{P}_{s_2}$	0.0192	0.9381	0.9424	0.0258	0.7161	0.7967
	$\mathbf{P}_{s_3}$	0.0186	0.9411	0.9468	0.0246	0.7426	0.8124
	SPS	0.0187	0.9402	0.9423	0.0245	0.7428	0.8125
NIR	$\mathbf{P}_{s_1}$	0.0312	0.8850	0.8905	0.0348	0.8581	0.8630
	$\mathbf{P}_{s_2}$	0.0318	0.8812	0.8841	0.0354	0.8501	0.8531
	$\mathbf{P}_{s_3}$	0.0313	0.8845	0.8900	0.0353	0.8586	0.8494
	SPS	0.0312	0.8849	0.8898	0.0349	0.8589	0.8597
SWIR1	$\mathbf{P}_{s_1}$	0.0357	0.9500	0.9514	0.0578	0.7720	0.7560
	$\mathbf{P}_{s_2}$	0.0347	0.9528	0.9532	0.0546	0.7799	0.7570
	$\mathbf{P}_{s_3}$	0.0339	0.9550	0.9563	0.0607	0.7342	0.7028
	SPS	0.0340	0.9547	0.9553	0.0548	0.7765	0.7572
SWIR2	$\mathbf{P}_{s_1}$	0.0281	0.9521	0.9546	0.0450	0.7486	0.7245
	$\mathbf{P}_{s_2}$	0.0269	0.9556	0.9574	0.0421	0.7667	0.7370
	$\mathbf{P}_{s_3}$	0.0263	0.9578	0.9600	0.0474	0.7292	0.6791
	SPS	0.0262	0.9579	0.9600	0.0425	0.7628	0.7318

TABLE VII

ACCURACY COMPARISON AMONG PSDFA USING DIFFERENT SIMILAR IMAGE PAIRS IN THE CIA AND LGC DATASETS

	CIA				LGC			
	$\mathbf{P}_{s_1}$	$\mathbf{P}_{s_2}$	$\mathbf{P}_{s_3}$	SPS	$\mathbf{P}_{s_1}$	$\mathbf{P}_{s_2}$	$\mathbf{P}_{s_3}$	SPS
RMSE	0.0210	0.0213	0.0211	0.0208	0.0215	0.0217	0.0212	0.0209
CC	0.9217	0.9190	0.9211	0.9224	0.8746	0.8769	0.8816	0.8852
SSIM	0.9416	0.9399	0.9413	0.9422	0.9102	0.9123	0.9121	0.9149
ERGAS	0.7341	0.7397	0.7334	0.7239	0.8525	0.8536	0.8313	0.8216
SAM	0.0679	0.0700	0.0683	0.0672	0.0872	0.0898	0.0869	0.0849

TABLE VIII

COMPARISON AMONG THE PSDFA USING AUP, APA, AND REMOVING UPSCALING IN THE CIA DATASET. BOLD INDICATES THE BETTER INDICES

	with AUP			with APA			Remove upscaling		
index	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
2	**0.0246**	0.8976	**0.9260**	0.0248	**0.8989**	0.9251	0.0248	0.8980	0.9248
3	**0.0216**	**0.9263**	**0.9470**	0.0218	0.9252	0.9459	**0.0216**	0.9261	0.9464
4	0.0207	**0.9238**	0.9486	**0.0205**	0.9232	0.9486	0.0208	0.9221	0.9479
5	**0.0213**	**0.9182**	**0.9423**	**0.0213**	0.9176	0.9176	**0.0213**	0.9171	0.9420
6	**0.0226**	0.9287	0.9416	**0.0226**	**0.9292**	**0.9418**	0.0232	0.9246	0.9379
7	**0.0272**	**0.8980**	**0.8997**	0.0273	0.8972	0.8993	0.0274	0.8968	0.8984
8	**0.0207**	**0.9403**	**0.9508**	0.0213	0.9388	0.9493	0.0212	0.9388	0.9494
9	**0.0232**	**0.9318**	**0.9413**	0.0247	0.9245	0.9352	0.0247	0.9247	0.9353
10	**0.0169**	**0.9725**	**0.9770**	0.0175	0.9704	0.9750	0.0174	0.9709	0.9754
11	**0.0127**	**0.9863**	**0.9870**	0.0131	0.9833	0.9847	0.0129	0.9842	0.9855
12	0.0201	0.8871	0.9304	**0.0200**	**0.8882**	**0.9313**	**0.0200**	0.8879	0.9312
13	**0.0198**	**0.8840**	**0.9247**	**0.0198**	0.8828	0.9243	**0.0198**	0.8824	0.9241
14	**0.0192**	**0.8961**	**0.9318**	**0.0192**	0.8955	0.9314	**0.0192**	0.8950	0.9313
average	**0.0208**	**0.9224**	**0.9422**	0.0211	0.9211	0.9411	0.0211	0.9207	0.9407

images in all pairs of the CIA dataset except the 12th and those of the fifth–11th pairs of the LGC dataset. In other pairs,

TABLE IX

COMPARISON AMONG THE PSDFA USING AUP, APA, AND REMOVING UPSCALING IN THE LGC DATASET. BOLD INDICATES THE BETTER INDICES

	with AUP			with APA			Remove upscaling		
index	RMSE	CC	SSIM	RMSE	CC	SSIM	RMSE	CC	SSIM
2	0.0191	**0.9171**	0.9311	**0.0189**	0.9170	**0.9328**	0.0190	0.9156	0.9318
3	0.0201	0.9076	0.9289	**0.0199**	**0.9081**	**0.9298**	0.0201	0.9076	0.9290
4	0.0233	0.9090	**0.9283**	0.0229	0.9137	0.9254	**0.0228**	**0.9149**	0.9276
5	**0.0170**	**0.8489**	**0.9150**	0.0178	0.8475	0.9101	0.0177	0.8481	0.9106
6	**0.0238**	**0.7827**	**0.8509**	0.0242	0.7791	0.8428	0.0240	0.7717	0.8447
7	**0.0322**	**0.7700**	**0.8073**	0.0327	**0.7700**	0.8067	0.0336	0.7583	0.7959
8	**0.0233**	**0.8759**	**0.9082**	0.0239	0.8693	0.9031	0.0237	0.8736	0.9050
9	**0.0189**	**0.9180**	**0.9428**	0.0201	0.9133	0.9393	0.0198	0.9159	0.9414
10	**0.0238**	**0.9147**	**0.9256**	**0.0238**	**0.9147**	0.9251	0.0239	0.9144	0.9249
11	**0.0145**	**0.9500**	**0.9644**	0.0152	0.9499	0.9642	0.0152	0.9496	0.9641
12	**0.0140**	0.9434	0.9615	**0.0140**	0.9430	0.9614	**0.0140**	**0.9435**	**0.9435**
average	**0.0209**	**0.8852**	**0.9149**	0.0212	0.8841	0.9128	0.0212	0.8830	0.9124

the AUP leads to slight impairment. Therefore, it can be said the AUP increases the fusion accuracy in general. As for the APA, it increases the fusion accuracy of the second–sixth and 12th–14th pairs of the CIA dataset, and that of the second, third, seventh, tenth, and 11th pairs of the LGC dataset, while the accuracy of other pairs is decreased. The APA also brings improvement for the fusion of both CIA and LGC datasets, which can be seen through the average indices. However, the improvement of APA is smaller than the AUP, which is due to the AUP simulates the real spatial response while the APA simplifies that. In summary, this comparison indicates the contribution and necessity of the AUP of PSDFA, which further improves the fusion accuracy.

C. Limitations of the PSDFA

Through the validation of two experiments in local computers and GEE, respectively, the excellent fusion performance of the PSDFA has been proven. After that, the effect of SPS and AUP, which are two key components of the PSDFA, are also investigated. In general, this article has exhibited the PSDFA comprehensively. However, there are still some limitations in this method, which is necessary to be pointed out. First, the SPS just considers three strategies for producing the synthetic image pairs presently. The PSDFA relies on SPS, for which more synthetic image pairs in SPS mean the bigger possibility to improve the performance of the PSDFA. In the future, more strategies will be adopted by the SPS. In addition, the SPS sometimes makes mistakes. Second, the AUP may bring a negative effect on the accuracy of the PSDFA, which can be seen from Tables VIII and IX. Though it is occasional, PSDFA cannot avoid this risk currently. This may be addressed via enhancing the AUP. In addition, the performance of the PSDFA in GEE is sensitive to the accuracy of cloud removal of fine images. A potential solution is using more powerful cloud removal approaches to process images before fusion of

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PSDFA, rather than just using their QA masks to removing the cloud. However, though a better cloud removal approach may alleviate this problem, improvement for PSDFA itself to get rid of that is necessary. In addition, this work provides a way to handle the band difference problem in spatiotemporal fusion; however, it does not explicitly point out how that affects the spatiotemporal fusion. That will be further explored in the future.

V. Conclusion

In this work, the band difference is considered in the spatiotemporal fusion of remote sensing imagery for the first time, based on which a new spatiotemporal fusion approach, i.e., the PSDFA, is developed. The PSDFA is capable of selecting the best fusion parameters for each band to ensure their fusion performance. In addition, the PSDFA can also be implemented in GEE through simplification; thus, it is a multiplatform fusion method. To test the PSDFA, two experiments are conducted in local computers and GEE, respectively. In local computers, the PSDFA is compared with five state-of-the-art spatiotemporal fusion methods, including the FSDAF, Fit-FC, RASDF, ESTARFM, and VIPSTF, on two public Landsat–MODIS datasets. The experimental results demonstrate that the PSDFA has outstanding qualitative and quantitative fusion performance for different bands. Plus, the PSDFA is balanced for different bands. In GEE, it is compared with another GEE-based spatiotemporal fusion approach, i.e., HISTARFM, in two areas of the USA. The fusion results indicate that the PSDFA is more robust and accurate than HISTARFM. However, the performance of PSDFA relies on the accuracy of the cloud flag of fine images in GEE; thus, it is recommended to use better cloud removal approaches to process the fine images when using the PSDFA in GEE. We believe that the PSDFA is a good choice for providing medium spatial resolution remote sensing image series in both local computers and GEE, to support the applications and researches on agriculture, ecology, environment, and so on. In the future, we will further enhance the PSDFA, via overcoming its flaws and combining it with gap filling strategy, to improve its applicability in more fusion scenarios.

References

[1] C. Liao, J. Wang, T. Dong, J. Shang, J. Liu, and Y. Song, "Using spatio-temporal fusion of Landsat-8 and MODIS data to derive phenology, biomass and yield estimates for corn and soybean," *Sci. Total Environ.*, vol. 650, pp. 1707–1721, Feb. 2019.

[2] E. Nduati, Y. Sofue, A. Matniyaz, J. Park, W. Yang, and A. Kondoh, "Cropland mapping using fusion of multi-sensor data in a complex urban/peri-urban area," *Remote Sens.*, vol. 11, no. 2, p. 207, Jan. 2019.

[3] K. R. Knipper et al., "Evapotranspiration estimates derived using thermal-based satellite remote sensing and data fusion for irrigation management in California vineyards," *Irrigation Sci.*, vol. 37, no. 3, pp. 431–449, May 2019.

[4] V. Heimhuber, M. G. Tulbure, and M. Broich, "Addressing spatio-temporal resolution constraints in Landsat and MODIS-based mapping of large-scale floodplain inundation dynamics," *Remote Sens. Environ.*, vol. 211, pp. 307–320, Jun. 2018.

[5] M. Chiesi et al., "Spatio-temporal fusion of NDVI data for simulating soil water content in heterogeneous Mediterranean areas," *Eur. J. Remote Sens.*, vol. 52, no. 1, pp. 88–95, Jan. 2019.

[6] Y. Pan, F. Shen, and X. Wei, "Fusion of Landsat-8/OLI and GOCI data for hourly mapping of suspended particulate matter at high spatial resolution: A case study in the Yangtze (Changjiang) estuary," *Remote Sens.*, vol. 10, no. 2, p. 158, 2018.

[7] X. Zhu, F. Cai, J. Tian, and K. A. Williams, "Spatiotemporal fusion of multisource remote sensing data: Literature survey, taxonomy, principles, applications, and future directions," *Remote Sens.*, vol. 10, no. 4, p. 527, 2018.

[8] Q. Wang and P. M. Atkinson, "Spatio-temporal fusion for daily Sentinel-2 images," *Remote Sens. Environ.*, vol. 204, pp. 31–42, Jan. 2018.

[9] M. Zhang, H. Lin, H. Sun, and Y. Cai, "Estimation of vegetation productivity using a Landsat 8 time series in a heavily urbanized area, central China," *Remote Sens.*, vol. 11, no. 2, p. 133, Jan. 2019.

[10] X. Zhou, P. Wang, K. Tansey, S. Zhang, H. Li, and L. Wang, "Developing a fused vegetation temperature condition index for drought monitoring at field scales using Sentinel-2 and MODIS imagery," *Comput. Electron. Agricult.*, vol. 168, Jan. 2020, Art. no. 105144.

[11] Y. Li, J. Li, L. He, J. Chen, and A. Plaza, "A new sensor bias-driven spatio-temporal fusion model based on convolutional neural networks," *Sci. China Inf. Sci.*, vol. 63, no. 4, 2020, Art. no. 140302.

[12] J. Li, Y. Li, L. He, J. Chen, and A. Plaza, "Spatio-temporal fusion for remote sensing data: An overview and new benchmark," *Sci. China Inf. Sci.*, vol. 63, no. 4, Apr. 2020, Art. no. 140301.

[13] B. Zhukov, D. Oertel, F. Lanzl, and G. Reinhackel, "Unmixing-based multisensor multiresolution image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 3, pp. 1212–1226, May 1999.

[14] F. Gao, J. G. Masek, M. R. Schwaller, and F. Hall, "On the blending of the Landsat and MODIS surface reflectance: Predicting daily Landsat surface reflectance," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 8, pp. 2207–2218, Jul. 2006.

[15] W. Zhang et al., "An enhanced spatial and temporal data fusion model for fusing Landsat and MODIS surface reflectance to generate high temporal Landsat-like data," *Remote Sens.*, vol. 5, no. 10, pp. 5346–5368, 2013.

[16] Y. Xu, B. Huang, Y. Xu, K. Cao, C. Guo, and D. Meng, "Spatial and temporal image fusion via regularized spatial unmixing," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 6, pp. 1362–1366, 2015.

[17] B. Huang and H. Zhang, "Spatio-temporal reflectance fusion via unmixing: Accounting for both phenological and land-cover changes," *Int. J. Remote Sens.*, vol. 35, no. 16, pp. 6213–6233, 2014.

[18] Y. Zhao, H. Bo, and H. Song, "A robust adaptive spatial and temporal image fusion model for complex land surface changes," *Remote Sens. Environ.*, vol. 208, pp. 42–62, Apr. 2018.

[19] Q. Cheng, H. Liu, H. Shen, P. Wu, and L. Zhang, "A spatial and temporal nonlocal filter-based data fusion method," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 8, pp. 4476–4488, Aug. 2017.

[20] X. Zhu, E. H. Helmer, F. Gao, D. Liu, J. Chen, and M. A. Lefsky, "A flexible spatiotemporal method for fusing satellite images with different resolutions," *Remote Sens. Environ.*, vol. 172, pp. 165–177, Jan. 2016.

[21] C. M. Gevaert and F. García-Haro, "A comparison of STARFM and an unmixing-based algorithm for Landsat and MODIS data fusion," *Remote Sens. Environ.*, vol. 156, pp. 34–44, Jan. 2015.

[22] J. Chen, L. Wang, R. Feng, P. Liu, W. Han, and X. Chen, "CycleGAN-STF: Spatiotemporal fusion via CycleGAN-based image generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5851–5865, Jul. 2021.

[23] W. Han et al., "Geological remote sensing interpretation using deep learning feature and an adaptive multisource data fusion network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[24] Y. Peng et al., "Spatiotemporal reflectance fusion via tensor sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[25] L. Li, P. Liu, J. Wu, L. Wang, and G. He, "Spatiotemporal remote-sensing image fusion with patch-group compressed sensing," *IEEE Access*, vol. 8, pp. 209199–209211, 2020.

[26] Y. Li, R. Cai, J. Li, Z. Liu, L. Meng, and L. He, "Pansharpening-based spatio-temporal fusion for predicting intense surface changes," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624114.

[27] W. Shi, D. Guo, and H. Zhang, "A reliable and adaptive spatiotemporal data fusion method for blending multi-spatiotemporal-resolution satellite images," *Remote Sens. Environ.*, vol. 268, Jan. 2022, Art. no. 112770.

[28] Q. Wang, Y. Tang, X. Tong, and P. M. Atkinson, "Virtual image pair-based spatio-temporal fusion," *Remote Sens. Environ.*, vol. 249, Nov. 2020, Art. no. 112009.

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[29] J. Li, Y. Li, R. Cai, L. He, J. Chen, and A. Plaza, "Enhanced spatiotemporal fusion via MODIS-like images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.
[30] H. Song, Q. Liu, G. Wang, R. Hang, and H. Bo, "Spatiotemporal satellite image fusion using deep convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 821–829, Feb. 2018.
[31] Z. Tan, P. Yue, L. Di, and J. Tang, "Deriving high spatiotemporal remote sensing images using deep convolutional network," *Remote Sens.*, vol. 10, p. 1066, Jul. 2018.
[32] X. Liu, C. Deng, J. Chanussot, D. Hong, and B. Zhao, "StfNet: A two-stream convolutional neural network for spatiotemporal image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6552–6564, Sep. 2019.
[33] Z. Tan, L. Di, M. Zhang, L. Guo, and M. Gao, "An enhanced deep convolutional model for spatiotemporal image fusion," *Remote Sens.*, vol. 11, no. 24, p. 2898, 2019.
[34] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, and R. Moore, "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sens. Environ.*, vol. 202, pp. 18–27, Dec. 2017.
[35] H. Tamiminia, B. Salehi, M. Mahdianpari, L. Quackenbush, S. Adeli, and B. Brisco, "Google Earth engine for geo-big data applications: A meta-analysis and systematic review," *ISPRS J. Photogramm. Remote Sens.*, vol. 164, pp. 152–170, Jun. 2020.
[36] Q. Wang, Y. Tang, and P. M. Atkinson, "The effect of the point spread function on downscaling continua," *ISPRS J. Photogramm. Remote Sens.*, vol. 168, no. 2, pp. 251–267, 2020.
[37] R. Fan, J. Li, W. Song, W. Han, J. Yan, and L. Wang, "Urban informal settlements classification via a transformer-based spatial–temporal fusion network using multimodal remote sensing and time-series human activity data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 111, Jul. 2022, Art. no. 102831.
[38] W. Han et al., "Improving training instance quality in aerial image object detection with a sampling-balance-based multistage network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10575–10589, Dec. 2021.
[39] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
[40] R. Fan, R. Feng, W. Han, and L. Wang, "Urban functional zone mapping with a bibranch neural network via fusing remote sensing and social sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11737–11749, 2021.
[41] X. Zhang et al., "Nearshore bathymetry based on ICESat-2 and multispectral images: Comparison between Sentinel-2, Landsat-8, and testing Gaofen-2," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2449–2462, 2022.
[42] I. V. Emelyanova, T. R. McVicar, T. G. Van Niel, L. T. Li, and A. I. van Dijk, "Assessing the accuracy of blending Landsat–MODIS surface reflectances in two landscapes with contrasting spatial and temporal dynamics: A framework for algorithm selection," *Remote Sens. Environ.*, vol. 133, pp. 193–209, Jun. 2013.
[43] Á. Moreno-Martínez et al., "Multispectral high resolution sensor fusion for smoothing and gap-filling in the cloud," *Remote Sens. Environ.*, vol. 247, Sep. 2020, Art. no. 111901.



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